

Andreas C. Kapourani

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Research and Professional Experience

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| Relation
2026–Present | Director of Data Science and Machine Learning.
Leading a team of machine learning researchers and data scientists developing DNA foundation models to connect human genetic variation with disease biology and therapeutic opportunities.
Driving R&D projects at the intersection of AI, genomics and biomedicine , translating scientific questions into machine learning programmes with clear milestones and deliverables.
Working closely with internal and external stakeholders to shape project strategy, coordinate cross-functional efforts, communicate scientific and technical progress, and move key programmes forward. |
| Relation
2023–2026 | Associate Director Data Science.
Built and developed sequence to function modelling approaches to link genetic variation with biological processes, complex traits and disease mechanisms.
Led a causal structure learning project to identify gene regulatory networks, and contributed to single-cell expression modelling with VAEs for predicting combinatorial genetic interventions.
Developed robust Bayesian models for analysing in-house experimental data, contributed to experimental design, and communicated findings to scientific and operational stakeholders. |
| The University of
Edinburgh
2018–2023 | Cross-disciplinary research fellow, MRC Human Genetics Unit and School of Informatics.
Advisors Dr Catalina Vallejos and Prof Neil Henderson
Project: Machine learning and statistical modelling of single-cell genomics
Developed ML models to elucidate relationship between tissue localization and gene expression variation and understand how location and function interact to regulate the transition from healthy to pathological disease states.
Analysed multi-modal single cell data , including sc/snRNA-seq, sequence- and image-based spatial transcriptomics, to decode molecular mechanisms regulating liver fibrosis and regeneration.
Developed latent class model to identify transcriptional subtypes in human-derived glioblastoma stem cell lines. |
| The University of
Edinburgh
2015–2021 | Teaching Support, School of Informatics.
Supervisor: Co-supervised MInf student for 2-year research project on cancer data science and epigenetics. Also supervised Erasmus student on machine learning for computational biology.
Tutor, demonstrator, marker for various courses at the School of Informatics. |
| DaXtra Technologies
2013–2014 | Junior Software Developer, Edinburgh.
Adapted company's proprietary software to client's needs: integration with 3rd party database management systems, web services and online job boards, including Indeed and LinkedIn. |

Education

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| The University of
Edinburgh
2015–2018 | PhD in Data Science, School of Informatics.
Under the supervision of Prof. Guido Sanguinetti
Thesis: Spatial statistical modelling of epigenomic variability.
Developed statistical machine learning methods to model (single-cell) genomics data and capture spatial correlations of epigenetic marks that would uncover the interplay between genetic and epigenetic mechanisms in transcriptional regulation.
Analysed single-cell multi-omics data to investigate regulatory relationships during mouse embryo development. |
| The University of
Edinburgh
2014–2015 | MSc by Research in Data Science with Distinction, School of Informatics.
Dissertation: Mixture modelling of high-throughput biomedical data. |

Developed machine learning approach to extract higher order methylation features that were predictive of gene expression levels.

Performed integrative clustering of heterogeneous sources of biological data.

The University of Edinburgh 2012–2013	MSc in Artificial Intelligence with Distinction , <i>School of Informatics</i> . Dissertation: Unsupervised motif discovery from acoustic time series data.
Harokopio University of Athens 2008–2012	BSc in Informatics with Distinction , <i>Department of Informatics and Telematics, Greece</i> . Dissertation: Integration of external applications in the Unity academic social network environment using Google's OpenSocial API.
Roskilde University 2010–2011	Erasmus Placement , <i>Department of Computer Science, Denmark</i> . Project: Developed <i>Find Friend Location</i> , a distributed real-time location tracking system for Android mobiles provided by SOAP Web Services.

Awards and Distinctions

2021	Nominated as supervisor of the year for the EUSA teaching awards at the University of Edinburgh.
2016	Best paper award at the 15 th European Conference on Computational Biology (ECCB).
2013	Winner of Travel App Prize (Skyscanner) University of Edinburgh (http://easyscn.appspot.com/). Natural language processing system for converting natural text to flight queries.
2012	Bodossaki foundation scholarship for postgraduate studies .
2009–2012	Scholarship from Greek state scholarships foundation (IKY) for excellent performance, highest ranked student for the 1 st , 2 nd , 3 rd and 4 th academic years.
2010	Erasmus placement scholarship for international studies.

Technical Skills

Machine Learning/ Statistics	Deep learning, Deep generative models, Generalized linear models, Bayesian inference, Gaussian processes, Approximate inference, Optimization, Latent variable models, Graphical models, Mixture models, Neural networks, Markov chain Monte Carlo, Kernel methods.
Languages	Python, PyTorch, Pyro, R, Stan, Java, C/C++, MATLAB, SQL, HTML.

Computational Genomics Software

SeuratPipe	SeuratPipe enables reproducible computational analysis of single-cell genomics (sc/snRNA-seq and Visium) by wrapping common tasks in easy to run pipelines (andreaskapou.github.io/SeuratPipe/).
scMET	scMET shares information across cells and genomic features using a Bayesian framework to quantify genuine biological single-cell epigenetic heterogeneity (bioconductor.org/packages/scMET/).
Melissa	Melissa jointly clusters and imputes single-cells by accounting for local correlations via a generalised linear model and global similarities using a mixture model (bioconductor.org/packages/Melissa/).
BPRMeth	BPRMeth extracts higher order methylation features that can be used for downstream modelling efforts, such as predicting gene expression or clustering genomic regions or cells according to their methylation patterns. (bioconductor.org/packages/BPRMeth/).

Public Engagement

Ongoing	Tutorial-like blog with accompanying code on statistical machine learning and Bayesian inference: https://andreaskapou.github.io/blog/ .
2020	Involved in the IGC statistical seminar series , teaching biomedical researchers how to use statistical and machine learning methods to answer research questions.
2018	Contributed to the XD Challenge public event: provided students with challenging biomedical problems and how they could apply computational approaches to tackle them.

List of Publications

- 2026 Regep, C., **Kapourani, C.A.**, Sofyali, E., Dobrowolska, A., Loukas, G., Anighoro, A., ... (2026). **A DNA foundation model predicts osteoporosis risk genes without proximity bias.** *bioRxiv*, preprint bioRxiv:2026.03.09.707383.
- 2025 Del Vecchio, A., **Kapourani, C.A.**, Athar, A.M., Dobrowolska, A., Anighoro, A., ... (2025). **PatchDNA: A flexible and biologically-informed alternative to tokenization for DNA.** *bioRxiv*, preprint bioRxiv:2025.11.28.691095.
- 2024 **Kapourani, C.A.**, Del Vecchio, A., Dobrowolska, A., Anighoro, A., Hessel, E.M., ... (2024). **Gene-centric evaluation of causal variant prediction for DNA models.** *ICML Workshop on Machine Learning for Life and Material Science: From Theory to Industry*.
- 2024 Matchett KP, Wilson-Kanamori J*, **Kapourani, C.A.***, Portman, JP*, Fercoq F, May S, Mackey J, Brice M, Zadjel E, Beltran-Sierra M, Sutherland E, Wilson G, Wallace SJ, Kitto L, Younger NT, Dobie R, Oniscu GC Wigmore SJ, Ramachandran P, Vallejos CA, Carragher NO, Simpson KJ, Kendall TJ, Acute Liver Failure Study Group; Rule JA, Lee WM, Hoare M, Weston CJ, Marioni JC, Teichmann ST, Bird TG, Carlin L, Henderson NC. **Multimodal decoding of human liver regeneration.** *Nature*, 630, 158–165
- 2024 Gaudet, T., Del Vecchio, A., Carrami, E. M., Cudini, J., **Kapourani, C.A.**, Uhler, C., & Edwards, L. (2024). **Season combinatorial intervention predictions with Salt & Peper.** *arXiv*, preprint arXiv:2404.16907.
- 2021 **Kapourani, C.A.***, Argelaguet, R*, Sanguinetti, G, Vallejos, C (2021). **scMET: Bayesian modelling of DNA methylation heterogeneity at single-cell resolution.** *Genome Biology*, 22(1), pp.1-21.
- 2021 Gangoso, E., Southgate B., Bradley L., Rus, S., Galvez-Cancino, F., McGivern, N., **Kapourani, C.A.**, Byron, A., Ferguson, K., ..., Sproul, D., Margaret Frame, M., Serrels, A. and Pollard, S. (2021). **Glioblastoma stem cells hijack myeloid-affiliated transcription factors via epigenetic immunoediting to elicit immune-evasion,** *Cell*, 184(9), pp.2454-2470.
- 2019 Argelaguet, R., Clark, S.J., Mohammed, H., Stapel, L.C., Krueger, C., **Kapourani, C.A.**, Imaz-Rosshandler, I., Lohoff, T., Xiang, Y., Hanna, C.W., Smallwood, S., Ibarra-Soria, X., Buettner, F., Sanguinetti, G., Xie, W., Krueger, F., Gottgens, B., Rugg-Gunn, P.J., Kelsey, G., Dean, W., Nichols, J., Stegle, O., Marioni J.C. and Reik, W. (2019). **Multi-omics profiling of mouse gastrulation at single-cell resolution.** *Nature*, 576(7787), 487-491.
- 2019 **Kapourani, C.A.** and Sanguinetti, G. (2019). **Melissa: Bayesian clustering and imputation of single-cell methylomes.** *Genome Biology*, 20(1), 1-15.
- 2018 Clark, S.J., Argelaguet, R., **Kapourani, C.A.**, Stubbs, T.M., Lee, H.J., Alda-Catalinas, C., Krueger, F., Sanguinetti, G., Kelsey, G., Marioni, J.C., Stegle, O. and Reik, W., (2018). **scNMT-seq enables joint profiling of chromatin accessibility DNA methylation and transcription in single cells.** *Nature Communications*, 9(1), p.781.
- 2018 **Kapourani, C.A.** and Sanguinetti, G. (2018). **BPRMeth: a flexible Bioconductor package for modelling methylation profiles.** *Bioinformatics*.
- 2016 **Kapourani, C.A.** and Sanguinetti, G. (2016). **Higher order methylation features for clustering and prediction in epigenomic studies.** *Bioinformatics*, 32 (17), i405-i412. (Best Paper Award in ECCB 2016)
- 2014 Hatzi, O., Meletakis, G., Katsivelis, P., **Kapourani, C.A.**, Nikolaidou, M. and Anagnostopoulos, D. (2014). **Extending the Social Network Interaction Model to Facilitate Collaboration through Service Provision.** In Enterprise, Business-Process and Information Systems Modeling (pp. 94-108). Springer Berlin Heidelberg.

List of Conference Presentations

- July 2024 Poster presentation at *ICML* conference. Title: Gene-centric evaluation of causal variant prediction for DNA large language models.
- Nov 2020 Oral presentation at *Biology of Genomes* meeting, Cold Spring Harbor Laboratory, New York, USA. Title: scMET: Bayesian modelling of DNA methylation heterogeneity at single-cell resolution.
- May 2020 Poster presentation at *Single cell biology* conference, Wellcome Genome Campus, Cambridge, UK. Title: scMET: Bayesian modelling of DNA methylation heterogeneity at single-cell resolution.

- May 2019 Oral presentation at *Centre for Genomic & Experimental Medicine* meeting, IGC, Edinburgh, UK. Title: Single-cell RNA-seq for identifying cell sub-population specific cis-eQTLs.
- March 2018 Oral presentation at *Mathematical and Statistical Aspects of Molecular Biology (MASAMB)* workshop, St Andrews, UK. Title: Melissa: Bayesian clustering and imputation of single-cell methylomes. DOI: [10.7490/f1000research.1115317.1](https://doi.org/10.7490/f1000research.1115317.1)
- July 2017 Poster presentations at *International Society for Computational Biology (ISMB)* conference, Prague, Czech Republic. Title: Bayesian hierarchical modelling of single-cell methylation profiles
- May 2017 Poster presentations at *Statistical Challenges in Single-Cell Biology* meeting, Ascona, Switzerland. Title: Bayesian hierarchical modelling of single-cell methylation profiles
- Sept 2016 Oral presentation at *15th European Conference on Computational Biology (ECCB)*, Hague, Netherlands. Title: Higher order methylation features for clustering and prediction in epigenomic studies. DOI: [10.7490/f1000research.1113102.1](https://doi.org/10.7490/f1000research.1113102.1)
- Aug 2015 Oral presentation at *Statistical Modeling of Epigenomics and Gene Regulation (SMEGR)* workshop, Harvard University, USA. Title: Mixture modelling of DNA methylation profiles.

Peer Review

Reviewer for peer-reviewed journals including *Genome Biology*, *Bioinformatics*, *PLOS Computational Biology*, *Statistical Applications in Genetics and Molecular Biology* and *IEEE/ACM Transactions on Computational Biology*.

Languages

Greek	Native Language
Albanian	Native Language
English	International English Language Testing System (IELTS)

References

Available on request